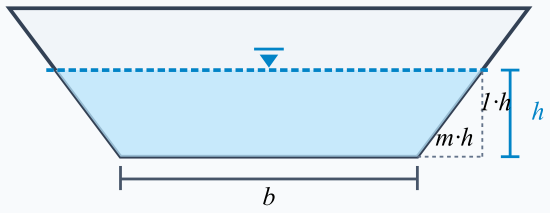
interpolate_h(Q, b, m_bank, S, **kwargs)

GAUCKLER-MANNING-STRICKLER & NEWTON-RAPHSON

1. TRAPEZOIDAL GEOMETRY



2. GEOMETRIC & ROUGHNESS FORMULAS

$$A(h) = h \cdot (b + m \cdot h)$$

$$P(h) = b + 2h \cdot \sqrt{(m^2 + 1)}$$

$$dA/dh = b + 2m \cdot h$$

$$dP/dh = 2 \cdot \sqrt{(m^2 + 1)}$$

$$k_{st} = 26 / (D_{90})^{1/6} \quad | \quad n_m = 1 / k_{st}$$

3. MANNING FORMULA & OBJECTIVE FUNCTION FORMULATION

$$Q_k(h) = (1 / n_m) \cdot A(h)^{5/3} \cdot P(h)^{-2/3} \cdot \sqrt{S}$$

$$\varepsilon = |Q - Q_k| / Q$$

$$F(h) = n_m \cdot Q \cdot P(h)^{2/3} - A(h)^{5/3} \cdot \sqrt{S} = 0$$

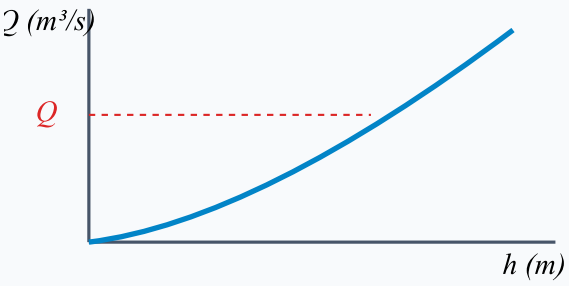
$$dF/dh = (2/3)\, n_m\, Q\, P^{-1/3}\, (dP/dh) - (5/3)\, A^{2/3}\, \sqrt{S}\, (dA/dh)$$

4. NEWTON–RAPHSON ALGORITHM SCHEME



k(h) *F(h), dF/dh, ε* $\varepsilon \leq 10^{-3}$ *?* **YES** *Return (h, ε, Q_k, k_{st})* **NO** $h_{n+1} = |h_n - F / F'|$ **iter = iter + 1**

5. Q VS. H CONVERGENCE CHARACTERISTIC



target h_n

6. VERIFICATION OUTPUT DATA

Parameter	Symbol	Value	Unit
Target Discharge	Q	10.50	m³/s
Bottom Width	b	5.00	m
Bank Slope	m	2.50	–
Bed Slope	S	0.0050	m/m
Grain Size	D_{90}	0.080	m
Strickler Coeff.	**k_{st}**	**39.65**	**m¹/³/s**
Computed Depth	**h_n**	**0.867**	**m**
Relative Error	**ε**	**1.4×10^{-4}**	**–**